E245 A2 2025

Question 1/*Vraag 1* [28]

(a) Explain why a reverse biased diode, for all practical purposes, does not conduct any current. Use sketches to help your explanation. Explain in terms of charge carrier movement both before and after the reverse bias voltage is applied.

(b) Determine the increase in forward bias voltage required to increase the conduction current by a factor of 20 for a silicon pn junction diode at a temperature of $300K$. Assume the diode is strongly biased in the forward direction.

(c) A diode is fabricated by doping intrinsic silicon with arsenic and gallium. Sketch the complete nonlinear diode I-V curve and the piecewise linear approximation on the same axes. Show both the forward bias and the reverse bias conditions. Indicate all important parameters on the sketch.

(d) Draw a circuit that would clamp an input voltage signal of 10 $\sin(\omega t)$ V to a minimum value of -2 V. Draw two cycles of the steady state input and output voltage waveforms versus time. Indicate any assumptions.

(e) For the circuit shown in Figure 1 below R_1=1 kΩ and R_2=10 kΩ. Plot v_o for $0\,\text{V} \leq v_I \leq 15\,\text{V}$. Assume V_γ=0.6 V and r_f=0 Ω. Show all calculations and indicate all breakpoint values on the curve.

(a) Verduidelik hoekom 'n trugespanne diode vir alle praktiese doeleindes geen stroom gelei nie. Gebruik sketse om te help met jou verduideliking. Verduidelik in opsigte van die beweging van ladingdraers voor en nadat die truspanning toegepas is. [4]

(b) Bepaal die toename in voorwaartse voorspanning wat benodig word om die geleidings stroom met 'n faktor van 20 te verhoog vir 'n silikon pn voegvlak diode by 'n temperatuur van $300K$. Neem aan dat die diode sterk in die voorwaartse rigting voorgespan is. [5]

(c) 'n Diode word vervaardig deur intrinsieke silikon met arseen en gallium te doteer. Skets die volledige nie-lineêre diode I-V-kromme en die stuksgewys-lineêre benadering op dieselfde asse. Toon beide die voorwaartse en die terugwaartse spannings toestande. Dui alle belangrike parameters op die skets aan. [5]

(d) Teken 'n stroombaan wat 'n intreespanning van 10 $\sin(\omega t)$ V tot 'n minimum van -2 V sal klem. Teken twee siklusse van die bestendige toestand intree- en uittreespannings teenoor tyd. Dui enige aannames aan. [5]

(e) Vir die stroombaan getoon in Figuur 1, R_1=1 kΩ en R_2=10 kΩ. Plot v_o vir $0\,\text{V} \leq v_i \leq 15\,\text{V}$. Aanvaar V_γ = 0.6 V en r_f=0 Ω. Toon al u bewerkings en dui alle breekpuntwaardes op die kenlyn aan. [9]

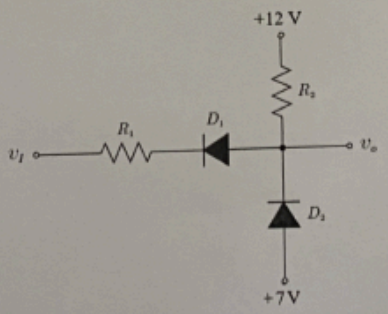


Figure 1: Multi-diode circuit/*Multi-diode stroombaan*

Question 2/*Vraag 2* [12]

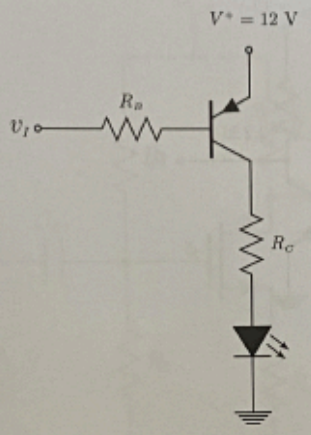


Figure 2: LED switching circuit/*LUD skakelbaan*

The switching circuit shown in Figure 2 is used to drive a light emitting diode (LED). For the transistor we have that β=100, $V_{EB(on)}$=0,7 V and $V_{EC(sat)}$=0,2 V. The LED functions optimally at a current of 10 mA and V_γ=1,8 V.

a) Calculate R_C and R_B. Make sure that the transistor is driven deep into saturation by choosing the input current double the minimum required value.

b) At what value for v_I will the transistor switch off?

c) Draw the load line sketch of the switch and show the static operating point (Q-point) when the transistor is in saturation. Also give numerical values for the Q-point and the intercepts on the axes.

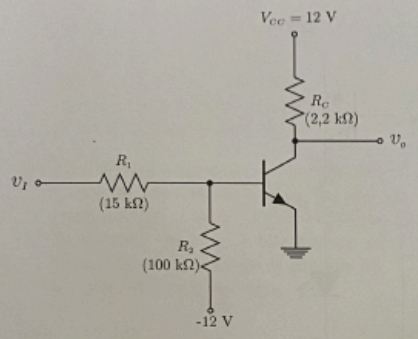
Die skakelbaan in Figuur 2 word gebruik om 'n liguitstraaldiode (LUD) aan te dryf. Vir die transistor geld dat β=100, $V_{EB(aan)}$=0,7 V en $V_{EC(sat)}$=0,2 V. Die LUD funksioneer optimaal by 'n stroom van 10 mA en V_γ=1,8 V.

a) Bepaal R_C en R_B. Maak seker dat die transistor diep in versadiging sal wees deur die intreestroom dubbel die minimum vereiste waarde te kies. [5]

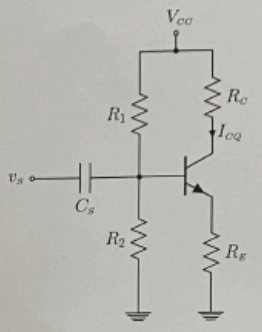
b) By watter waarde van v_I sal die transistor afskakel? [1]

c) Teken die laslynskets van die skakelaar en toon die statiese werkpunt (Q-punt) wanneer die transistor in versadiging is. Gee ook die numeriese waardes by die Q-punt en van die afsnitte op die asse. [6]

Question 3/*Vraag 3* [20]



(a)



(b)

Figure 3: Common-emitter amplifier circuits/*Gemene-emittor versterkerbane*

a) For the transistor in the circuit shown in Figure 3(a) we have that β=80, $V_{BE(on)}$=0,7 V and $V_{CE(sat)}$=0,2 V. Sketch the transfer curve of the circuit (v_o versus v_I) when $0 \leq v_I \leq 5$ V.

Tip: Consider using Thevenin!

b) A transistor circuit is shown in Figure 3(b). Show that, for typical bipolar transistor parameters, if $R_1||R_2 \approx 0.1\beta R_E$, the collector current becomes relatively insensitive to changes in β.

a) Die transistorparameters van die baan wat in Figuur 3(a) getoon word, word gegee deur β=80, $V_{BE(aan)}$=0,7 V en $V_{CE(sat)}$=0,2 V. Skets die oordragskenlyn van die baan (v_o teenoor v_I) wanneer $0 \leq v_I \leq 5$ V. [10]

Wenk: Oorweeg om Thevenin te gebruik!

b) 'n Transistorbaan word in die Figuur 3(b) getoon. Toon aan dat, vir tipiese bipolêre transistor parameters, as $R_1||R_2 \approx 0.1\beta R_E$, die kollektorstroom relatief onafhanklik raak van veranderinge in β. [10]

Question 4/*Vraag 4*

[20]

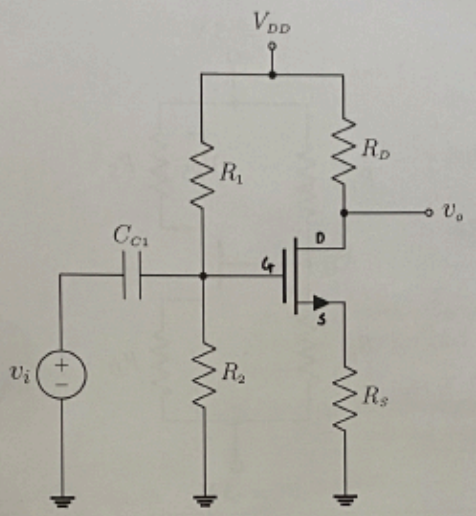


Figure 4: Common-source amplifier/*Gemene-bron versterker*

For the NMOS common-source amplifier shown in Figure 4, the transistor parameters are V_{TN}=0,8 V, K_n=1 mA/V^2, and λ=0. The circuit parameters are V_{DD}=5 V, R_S=1 kΩ, R_D=4 kΩ, and R_2=175 kΩ.

a) Draw the load line sketch for the amplifier.

b) Determine the values V_{DSQ}, I_{DQ} and V_{GSQ} so that the Q-point will be in the middle of the undistorted working range for the amplifier, i.e. design for optimum output voltage swing. Calculate the value for R_1 that will set the Q-point at the determined values.

Die transistor se parameters vir die getoonde gemene-bron NMOS versterker (Figuur 4) word gegee deur V_{TN}=0,8 V, K_n=1 mA/V^2 en λ=0. Die baan se parameters is V_{DD}=5 V, R_S=1 kΩ, R_D=4 kΩ, en R_2=175 kΩ.

a) Teken die laslynskets vir die getoonde versterker. [6]

b) Bepaal die waardes V_{DSQ}, I_{DQ} en V_{GSQ} sodat die Q-punt in die middel van die onvervormde werkbereik van die versterker sal wees, d.w.s. ontwerp vir optimale uittreeswaai. Bereken die waarde van R_1 wat die Q-punt by die bepaalde waardes sal vasmaak. [14]

Question 5/*Vraag 5*

[10]

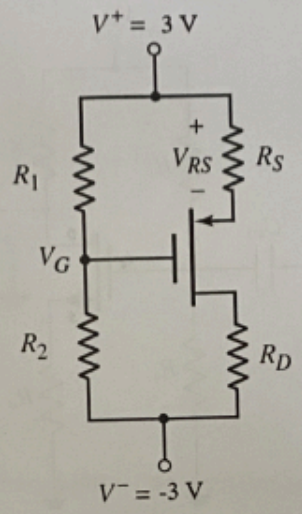


Figure 5: MOSFET amplifier/*MOSVET versterker*

In the circuit shown in Figure 5, R_1=7,5 kΩ, R_2=15 kΩ, R_S=500 Ω and R_D=2,2 kΩ. The transistor parameters are given as V_{TP}=–1,2 V and K_p=0,75 mA/V^2.

(a) Calculate I_{DQ}, V_{SGQ} and V_{SDQ}.

Vir die baan wat in Figuur 5 getoon word, geld dat R_1=7,5 kΩ, R_2=15 kΩ, R_S=500 Ω en R_D=2,2 kΩ. Die transistorparameters word gegee deur V_{TP}=–1,2 V en K_p=0,75 mA/V^2.

(a) Bepaal I_{DQ}, V_{SGQ} en V_{SDQ}. **[10]**